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BATTERY STRUCTURE AND ELECTRONIC DEVICE COMPRISING BATTERY STRUCTURE

Abstract

An electronic device is provided. The electronic device includes a battery cell, a battery pouch for sealing the battery cell, the battery pouch including an accommodation portion in which the battery cell is disposed and a first sealing portion that is bent to face the battery cell and including at least one groove, an electrode terminal including a first electrode tab connected to a positive plate of the battery cell and drawn out to the first sealing portion and a second electrode tab connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion, and a protection circuit module which includes at least one electronic element, and disposed on the first sealing portion such that the electrode element is positioned in the groove of the first sealing portion, wherein the protection circuit module is connected to the electrode terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/019660, filed on Dec. 1, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0174738, filed on Dec. 14, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0002844, filed on Jan. 9, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a battery structure and an electronic device including the same.

2. Description of Related Art

[0003] As the demand for portable electronic devices increases, the demand for batteries supplying power to electronic devices is also increasing. In keeping with this growing demand, technology in the battery field has also been steadily advancing.

[0004] Meanwhile, as the demand for miniaturization, slimming, and/or weight reduction of electronic devices increases, the disposition space for electronic elements is being reduced. For example, the disposition space for batteries within electronic devices is being reduced.

[0005] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0006] As electronic devices become miniaturized and slimmed down, the space occupied by a battery structure within an electronic device may be designed to be limited. In particular, electronic devices worn on a portion of a user's body, such as wearable electronic devices, may have size constraints to consider user comfort. As a result, the space occupied by a battery structure within a wearable electronic device may be reduced. Meanwhile, when implementing a battery structure in a limited space, the size of the battery cell may be limited. In this case, the capacity of the battery structure may be limited. As a result, consumer usage time for electronic devices may decrease.

[0007] According to an embodiment of the disclosure, the space occupied by a protection circuit module within the space allocated to a battery structure in an electronic device may be reduced. In this case, the size of the battery cell may be increased by the reduced space to ensure the capacity of the battery cell.

[0008] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a battery structure and an electronic device including the same.

[0009] Additional aspects will be set forth in part in the description which follows and, in part, will

be apparent from the description, or may be learned by practice of the presented embodiments.

[0010] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a battery cell, a battery pouch sealing the battery cell, the battery pouch including an accommodation portion in which the battery cell is disposed and a first sealing portion that is bent to face the battery cell and includes at least one groove, an electrode terminal including a first electrode tab connected to a positive electrode plate of the battery cell and drawn out to the first sealing portion and a second electrode tab connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion, and a protection circuit module including at least one electronic element and disposed on the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion, wherein the protection circuit module is connected to the electrode terminal.

[0011] In accordance with another aspect of the disclosure, a battery structure is provided. The battery structure includes a battery cell, a battery pouch sealing the battery cell, the battery pouch including an accommodation portion in which the battery cell is disposed and a first sealing portion that is bent to face the battery cell and includes at least one groove, an electrode terminal including a first electrode tab connected to a positive electrode plate of the battery cell and drawn out to the first sealing portion and a second electrode tab connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion, and a protection circuit module including at least one electronic element and disposed on the first sealing portion such that the electronic element is positioned in the groove of the first sealing portion, wherein the protection circuit module is connected to the electrode terminal.

[0012] According to an embodiment of the disclosure, a space between components constituting the battery structure may be reduced. In this case, the size of the battery cell may be increased by the reduced space between the components to ensure the capacity of the battery cell. For example, in the battery structure, the space occupied by the protection circuit module (PCR) of the battery structure may be reduced, and the size of the battery cell may be increased by the reduced space.

[0013] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure;

[0016] FIG. 2 is a block diagram illustrating a power management module and a battery according to an embodiment of the disclosure;

[0017] FIG. 3 is a view illustrating a battery structure in a state before being bent at a first sealing portion according to an embodiment of the disclosure;

[0018] FIG. 4A is a view illustrating a battery structure in a state in which a bent shape is formed at the first sealing portion according to an embodiment of the disclosure;

[0019] FIG. 4B is an enlarged view illustrating the first sealing portion of FIG. 4A according to an embodiment of the disclosure;

[0020] FIG. 4C is a rear perspective view illustrating the battery structure of FIG. 4A according to an embodiment of the disclosure;

[0021] FIG. 5A is a front view illustrating a state in which the first sealing portion of FIG. 4A is

folded to face the accommodation portion of the battery pouch according to an embodiment of the disclosure;

[0022] FIG. 5B is an enlarged view illustrating the first sealing portion of FIG. 5A according to an embodiment of the disclosure;

[0023] FIG. 6A is a perspective view illustrating a state in which a protection circuit module is disposed on the first sealing portion of the battery pouch according to an embodiment of the disclosure;

[0024] FIG. 6B is a front view illustrating a state in which the protection circuit module is disposed on the first sealing portion of the battery pouch according to an embodiment of the disclosure; and

[0025] FIGS. 7A and 7B illustrate views for comparing and describing the battery structure of the disclosure with a battery structure according to an embodiment of the disclosure.

[0026] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

[0027] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0028] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0029] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0030] In relation to the description of the drawings, similar reference numerals may be used for similar or related components.

[0031] In this document, each of phrases, such as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C” may include one of the items listed together in the corresponding one of the phrases, or all possible combinations thereof. Terms, such as “1.sup.st,” “2.sup.nd,” “first,” and “second” may be used to simply distinguish a given component from other corresponding components and do not limit the corresponding components in other aspects (e.g., importance or order). When a certain (e.g., a first) component is mentioned as being “coupled” or “connected” to another (e.g., a second) component, with or without a term “functionally” or “communicatively,” it means that the certain component may be connected to the other component directly (e.g., wiredly), wirelessly, or via a third component.

[0032] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include computer-executable instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0033] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g., a central

processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphical processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless-fidelity (Wi-Fi) chip, a Bluetooth™ chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a fingerprint sensor controller, a display drive integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0034] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

[0035] Referring to FIG. 1, an electronic device **101** in a network environment **100** may communicate with an external electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an external electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment of the disclosure, the electronic device **101** may communicate with the external electronic device **104** via the server **108**. According to an embodiment of the disclosure, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments of the disclosure, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments of the disclosure, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0036] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment of the disclosure, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment of the disclosure, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0037] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., a sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., the

neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0038] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0039] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0040] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0041] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment of the disclosure, the receiver may be implemented as separate from, or as part of the speaker.

[0042] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment of the disclosure, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0043] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment of the disclosure, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0044] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment of the disclosure, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0045] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment of the disclosure, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0046] A connecting terminal **178** may include a connector via which the electronic device **101**

may be physically connected with the external electronic device (e.g., the external electronic device **102**). According to an embodiment of the disclosure, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0047] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment of the disclosure, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0048] The camera module **180** may capture a still image or moving images. According to an embodiment of the disclosure, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0049] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment of the disclosure, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0050] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment of the disclosure, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0051] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the external electronic device **102**, the external electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment of the disclosure, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0052] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device

101, an external electronic device (e.g., the external electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment of the disclosure, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0053] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment of the disclosure, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment of the disclosure, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment of the disclosure, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**. According to various embodiments of the disclosure, the antenna module **197** may form a mm Wave antenna module. According to an embodiment of the disclosure, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0054] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0055] According to an embodiment of the disclosure, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment of the disclosure, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102** or **104**, or the server **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment of the disclosure, the external electronic device **104** may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural

network. According to an embodiment of the disclosure, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., a smart home, a smart city, a smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0056] FIG. **2** is a block diagram **200** illustrating the power management module **188** and the battery **189** according to an embodiment of the disclosure.

[0057] Referring to FIG. **2**, the power management module **188** may include charging circuitry **210**, a power adjuster **220**, or a power gauge **230**. The charging circuitry **210** may charge the battery **189** by using power supplied from an external power source outside the electronic device **101**. According to an embodiment of the disclosure, the charging circuitry **210** may select a charging scheme (e.g., normal charging or quick charging) based at least in part on a type of the external power source (e.g., a power outlet, a USB, or wireless charging), magnitude of power supplyable from the external power source (e.g., about 20 Watt or more), or an attribute of the battery **189**, and may charge the battery **189** using the selected charging scheme. The external power source may be connected with the electronic device **101**, for example, directly via the connecting terminal **178** or wirelessly via the antenna module **197**.

[0058] The power adjuster **220** may generate a plurality of powers having different voltage levels or different current levels by adjusting a voltage level or a current level of the power supplied from the external power source or the battery **189**. The power adjuster **220** may adjust the voltage level or the current level of the power supplied from the external power source or the battery **189** into a different voltage level or current level appropriate for each of some of the components included in the electronic device **101**. According to an embodiment of the disclosure, the power adjuster **220** may be implemented in the form of a low drop out (LDO) regulator or a switching regulator. The power gauge **230** may measure use state information about the battery **189** (e.g., a capacity, a number of times of charging or discharging, a voltage, or a temperature of the battery **189**).

[0059] The power management module **188** may determine, using, for example, the charging circuitry **210**, the power adjuster **220**, or the power gauge **230**, charging state information (e.g., lifetime, over voltage, low voltage, over current, over charge, over discharge, overheat, short, or swelling) related to the charging of the battery **189** based at least in part on the measured use state information about the battery **189**. The power management module **188** may determine whether the state of the battery **189** is normal or abnormal based at least in part on the determined charging state information. If the state of the battery **189** is determined to abnormal, the power management module **188** may adjust the charging of the battery **189** (e.g., reduce the charging current or voltage, or stop the charging). According to an embodiment of the disclosure, at least some of the functions of the power management module **188** may be performed by an external control device (e.g., the processor **120**).

[0060] The battery **189**, according to an embodiment of the disclosure, may include a protection circuit module (PCM) **240**. The PCM **240** may perform one or more of various functions (e.g., a pre-cutoff function) to prevent a performance deterioration of, or a damage to, the battery **189**. The PCM **240**, additionally or alternatively, may be configured as at least part of a battery management system (BMS) capable of performing various functions including cell balancing, measurement of battery capacity, count of a number of charging or discharging, measurement of temperature, or measurement of voltage.

[0061] According to an embodiment of the disclosure, at least part of the charging state information or use state information regarding the battery **189** may be measured using a corresponding sensor (e.g., a temperature sensor) of the sensor module **176**, the power gauge **230**, or the power management module **188** According to an embodiment of the disclosure, the corresponding sensor (e.g., a temperature sensor) of the sensor module **176** may be included as part of the PCM **240**, or may be disposed near the battery **189** as a separate device.

[0062] FIG. **3** is a view illustrating a battery structure in a state before being bent at a first sealing

portion according to an embodiment of the disclosure.

[0063] FIG. 4A is a view illustrating a battery structure in a state in which a bent shape is formed at the first sealing portion according to an embodiment of the disclosure.

[0064] FIG. 4B is an enlarged view illustrating the first sealing portion of FIG. 4A according to an embodiment of the disclosure.

[0065] FIG. 4C is a rear perspective view illustrating the battery structure of FIG. 4A according to an embodiment of the disclosure.

[0066] FIG. 5A is a front view illustrating a state in which the first sealing portion of FIG. 4A is folded to face the accommodation portion of the battery pouch according to an embodiment of the disclosure.

[0067] FIG. 5B is an enlarged view illustrating the first sealing portion of FIG. 5A according to an embodiment of the disclosure.

[0068] FIG. 6A is a perspective view illustrating a state in which a protection circuit module is disposed on the first sealing portion of the battery pouch according to an embodiment of the disclosure.

[0069] FIG. 6B is a front view illustrating a state in which the protection circuit module is disposed on the first sealing portion of the battery pouch according to an embodiment of the disclosure.

[0070] In the following description, components that are identical or similar to those described above are the same reference numerals will be using the same reference numerals.

[0071] According to an embodiment of the disclosure, a battery structure **300** (e.g., the battery **189** in FIG. 1) may include a battery cell **310**, a battery pouch **320**, an electrode terminal **360** including a first electrode tab **361** connected to a positive electrode plate of the battery cell **310** and a second electrode tab **362** connected to a negative electrode plate of the battery cell **310**, and a protection circuit module **400** that is in close contact with a portion of the battery pouch **320** to be connected to the electrode terminal **360**.

[0072] According to an embodiment of the disclosure, an electronic device (e.g., the electronic device **101** in FIG. 1) may include a battery structure **300** (e.g., the battery **189** of FIG. 1) that includes a battery cell **310** configured to supply power to various electronic elements included in the electronic device (e.g., a processor (e.g., the processor **120** in FIG. 1), memory (e.g., the memory **130** in FIG. 1), or a display (e.g., the display module **160** of FIG. 1)). The battery structure **300** may be, for example, a lithium-ion battery (Li-ion battery). The lithium-ion battery is a rechargeable battery that is chargeable using an external power source. The lithium-ion battery is widely used in mobile electronic devices due to the high energy density, excellent retention, and long life cycle thereof. In addition, a lithium ion polymer battery, which uses a solid or gel-type electrolyte, has a significantly lower risk of electrolyte leakage compared to other batteries using liquid electrolytes. The battery structure **300** may be a lithium-ion battery with high energy storage per unit volume but is not limited thereto.

[0073] According to an embodiment of the disclosure, the battery cell **310** may include a positive electrode plate, a negative electrode plate, and a separator. The positive electrode plate may function as a cathode during the discharge of the battery cell **310**, thereby absorbing lithium ions from the electrolyte contained in the battery cell **310** to form a lithium compound, and may function as an anode during charging, thereby releasing lithium ions. The positive electrode plate may include a positive electrode current collector including a thin metal plate, such as aluminum and/or a positive electrode active material. The positive electrode active material may include lithium compounds, for example, lithium cobalt oxide ($\text{Li}_{0.5}\text{CoO}_2$), lithium nickel oxide ($\text{Li}_{0.5}\text{NiO}_2$), lithium nickel cobalt oxide ($\text{Li}_{0.5}(\text{NiCo})\text{O}_2$), lithium nickel cobalt manganese oxide ($\text{Li}_{0.5}(\text{NiCoMn})\text{O}_2$), lithium nickel cobalt aluminum oxide ($\text{Li}_{0.5}(\text{NiCoAl})\text{O}_2$), spinel-type lithium manganese oxide ($\text{Li}_{0.5}\text{Mn}_{0.2}\text{O}_{0.4}$), manganese dioxide ($\text{MnO}_{0.5}$), and/or olivine-type compounds, such as lithium iron phosphate ($\text{Li}_{0.5}\text{FePO}_4$) and lithium manganese phosphate ($\text{Li}_{0.5}\text{MnPO}_4$). The negative

electrode plate may function as an anode during the discharge of the battery cell **310**, thereby releasing lithium ions into the electrolyte contained in the battery cell **310**, and may function as a cathode during charging, thereby storing reduced lithium atoms. The negative electrode plate may include a negative electrode current collector including a thin metal plate, such as copper and/or a negative electrode active material. The negative electrode active material may include, for example, artificial and/or natural graphite that stores lithium through intercalation, various silicon-based compounds, such as silicon oxide, silicon nitride and/or silicon carbide, and lithium metal or its various alloys and metal compounds that store metallic lithium. The separator may be a member that allows lithium ions to pass therethrough while preventing electrical contact between the positive electrode and the negative electrode. In an embodiment of the disclosure, the separator may include a polymer material, such as polyethylene or polypropylene, which has micropores of a size that allows lithium ion passage. The minimum unit of a battery including the negative electrode, the positive electrode, and the separator may be referred to as a battery cell **310** or an electrode assembly.

[0074] According to an embodiment of the disclosure, the battery structure **300** may include various types of batteries. In an embodiment of the disclosure, the battery structure **300** may include a jelly roll-type battery. The jelly roll may include a structure in which the positive electrode plate and the negative electrode plate are stacked and wound together. The positive electrode plate and the negative electrode plate may be wound together with a separator interposed therebetween. According to another embodiment of the disclosure, the battery structure **300** may include a stack-type electrode assembly in which the positive electrode plate, the negative electrode plate, and/or the separator are stacked. The positive electrode plate may refer to an electrode from which electrons flow out. Since a chemical reaction in which electrons are lost occurs at the positive electrode plate, the positive electrode plate may be understood as an electrode where an oxidation reaction takes place. A positive electrode active material for positive electrode activation may be coated on at least one surface of the positive electrode plate. The negative electrode plate may refer to an electrode into which electrons flow. Since a chemical reaction in which electrons are gained occurs at the negative electrode plate, the negative electrode plate may be understood as an electrode where a reduction reaction takes place. A negative electrode active material for negative electrode activation may be coated on at least one surface of the negative electrode plate. As such, a current may be generated through the redox reaction between the positive electrode plate and the negative electrode plate with the separator interposed therebetween.

[0075] According to an embodiment of the disclosure, the separator disposed between the positive electrode plate and the negative electrode plate may function as an electrolyte in the battery cell **310**. The separator may be made of a porous polymer material. For example, the separator may be made of a porous polyolefin film, polyvinylidene fluoride-hexafluoropropylene, polyvinylidene fluoride-trichloroethylene, polymethyl methacrylate, polyacrylonitrile, polyvinylpyrrolidone, polyvinyl acetate, ethylene vinyl acetate copolymer, polyethylene oxide, cellulose acetate, cellulose acetate butyrate, cellulose acetate propionate, cyanoethyl pullulan, cyanoethyl polyvinyl alcohol, cyanoethyl cellulose, cyanoethyl sucrose, pullulan, carboxymethyl cellulose, acrylonitrile-styrene-butadiene copolymer, polyimide, polyethylene terephthalate, polybutylene terephthalate, polyester, polyacetal, polyamide, polyether ether ketone, polyether sulfone, polyphenylene oxide, polyphenylene sulfide, polyethylene naphthalate, a nonwoven fabric film, a film having a porous web structure, or a mixture thereof. Inorganic particles may be bonded to one or both surfaces of the separator.

[0076] According to an embodiment of the disclosure, the positive electrode plate of the battery cell **310** may be connected to the first electrode tab **361**. For example, the first electrode tab **361** may be in contact with the positive electrode plate. In some embodiments, the first electrode tab **361** may be integrated with the positive electrode plate. The negative electrode plate of the battery cell **310** may be connected to the second electrode tab **362**, which will be described later. For example, the

second electrode tab **362** may be in contact with the negative electrode plate. In some embodiments, the second electrode tab **362** may be integrated with the negative electrode plate. [0077] According to an embodiment of the disclosure, as illustrated in FIG. 3, the battery pouch **320** may seal the battery cell **310**. In an embodiment of the disclosure, the battery pouch **320** may accommodate the positive electrode plate, the negative electrode plate, and the separator, and may seal the positive and negative electroplates and the separator from the external environment. For example, the battery pouch **320** may include a metal (e.g., aluminum or stainless steel), a polymer material, and/or a composite or laminate thereof. In an embodiment of the disclosure, an organic electrolyte (not illustrated) may be injected into and sealed within the battery pouch **320**. For example, the organic electrolyte may include a lithium salt and an organic solvent.

[0078] In an embodiment of the disclosure, referring to FIG. 3, the battery pouch **320** may include an accommodating portion **321**, in which the battery cell **310** is positioned, and sealing portions **330**, **340**, and **350**, which are residual areas remaining after enclosing the battery cell **310**. The accommodating portion **321** may be an inner space of the battery pouch **320** in which the battery cell **310** is disposed. The sealing portions **330**, **340**, and **350** may be areas where the remaining areas after accommodating the battery cell **310** are bonded. For example, the battery pouch **320** may seal the battery cell **310** as the sealing portions **330**, **340**, and **350** are bonded through heat sealing or an adhesive member. In an embodiment of the disclosure, the sealing portions **330**, **340**, and **350** may be formed on the left, right, and lower sides of the battery cell **310**. For example, the battery pouch **320** may include a first sealing portion **330** positioned on the lower side of the battery cell **310** (e.g., in the $-y$ direction based on FIG. 3), a second sealing portion **340** positioned on the right side of the battery cell **310** (e.g., in the $+x$ direction based on FIG. 3, first direction), and a third sealing portion **350** positioned on the left side of the battery cell **310** (e.g., in the $-x$ direction based on FIG. 3, second direction). In some embodiments, the sealing portions **330**, **340**, and **350** may be formed on the left side (e.g., in the $-x$ direction based on FIG. 3), right side (e.g., in the $+x$ direction based on FIG. 3), upper side (e.g., in the $+y$ direction based on FIG. 3), and lower side (e.g., in the $-y$ direction based on FIG. 3) of the battery cell **310**. In addition, the sealing portions **330**, **340**, and **350** may be formed in various positions relative to the battery cell **310** depending on the size and shape of the battery cell **310**.

[0079] According to an embodiment of the disclosure, as illustrated in FIGS. 4A, 4B, and 4C, the battery structure **300** may include a first electrode tab **361** connected to the positive electrode plate of the battery cell **310** and a second electrode tab **362** connected to the negative electrode plate. The first electrode tab **361** and the second electrode tab **362** may be drawn out to at least one of the sealing portions **330**, **340**, and **350**. For example, the first electrode tab **361** may be connected to the positive electrode plate of the battery cell **310** and drawn out to the first sealing portion **330**, and the second electrode tab **362** may be connected to the negative electrode plate of the battery cell **310** and drawn out to the first sealing portion **330**. Hereinafter, it is assumed that the first electrode tab **361** and the second electrode tab **362** are drawn out to the first sealing portion **330**.

[0080] According to an embodiment of the disclosure, the battery structure **300** may include a protection circuit module **400** (e.g., the battery protection circuit (the protection circuit module (PCM)) **240** in FIG. 2). The protection circuit module **400** may perform one or more of various functions (e.g., a pre-cutoff function) to prevent performance degradation or damage to the battery cell **310**. In an embodiment of the disclosure, the protection circuit module **400** may protect the battery cell **310** by blocking overcharging, overdischarging, or overcurrent.

[0081] According to an embodiment of the disclosure, as illustrated in FIG. 6A, the protection circuit module **400** may include a printed circuit board **410**, at least one electronic element **420** disposed on the printed circuit board **410**, and/or a connection terminal **430** disposed on the printed circuit board **410** and connected to the electrode terminal **360**. In an embodiment of the disclosure, the electronic element **420** may be an integrated circuit (IC) chip of the protection circuit module **400**. In an embodiment of the disclosure, the electronic element **420** may include a first electronic

element **421** and a second electronic element **422**. The first electronic element **421** may be disposed on a first surface **410A** of the printed circuit board **410**. The second electronic element **422** may be disposed on a second surface **410B** of the printed circuit board **410**. In an embodiment of the disclosure, referring to FIG. **6B**, the first surface **410A** of the printed circuit board **410** may be a surface facing the first surface **330A** of the first sealing portion **330**. The second surface **410B** of the printed circuit board **410** may be the opposite surface of the first surface **410A**, and may be a surface facing a counterpart of the electronic device. The first connection terminal **431** and the second connection terminal **432** may be disposed on the second surface **410B** of the printed circuit board **410**. In an embodiment of the disclosure, the connection terminal **430** may include a first connection terminal **431** connected to the first electrode tab **361** and a second connection terminal **432** connected to the second electrode tab **362**. Referring to FIGS. **6A** and **6B** to be described later, the first connection terminal **431** may be disposed on the second surface **410B** of the printed circuit board **410** to be electrically connected to the first electrode tab **361**. The second connection terminal **432** may be disposed on the second surface **410B** of the printed circuit board **410** to be electrically connected to the second electrode tab **362**.

[0082] In an embodiment of the disclosure, the protection circuit module **400** may be electrically connected to the battery cell **310**. In an embodiment of the disclosure, the protection circuit module **400** may be disposed adjacent to the first sealing portion **330**. For example, the printed circuit board **410** of the protection circuit module **400** may be disposed on and in close with a surface of the first sealing portion **330** (e.g., the first surface **330A** in FIG. **5B**) or the second surface (e.g., the second surface **330B** in FIG. **4B**). Referring to FIG. **6A**, at least a section of the first electrode tab **361** of the battery cell **310** may be bent to be connected to the first connection terminal **431** of the protection circuit module **400**. Likewise, at least a section of the second electrode tab **362** of the battery cell **310** may be bent to be connected to the second connection terminal **432** of the protection circuit module **400**.

[0083] According to an embodiment of the disclosure, as electronic devices become miniaturized and slimmed, the space occupied by the battery structure **300** within the electronic device may be designed to be limited. In particular, electronic devices worn on a portion of a user's body, such as wearable electronic devices, may have size constraints considering user comfort and weight perception. Accordingly, the space occupied by the battery structure **300** within a wearable electronic device may be reduced. Meanwhile, when implementing the battery structure **300** in a limited space, the size of the battery cell **310** may be limited. In this case, the capacity of the battery structure **300** may be limited. According to an embodiment of the disclosure, the space occupied by the protection circuit module **400** and/or the sealing portions **330** and **340** within the space occupied by the battery structure **300** in the electronic device may be reduced. In this case, the size of the battery cell **310** may be increased by the reduced space, thereby securing the capacity of the battery cell **310**. Hereinafter, a structure for reducing the space occupied by the protection circuit module **400** in the disclosure will be described below.

[0084] In an embodiment of the disclosure, referring to FIGS. **4A** and **4B**, the second sealing portion **340** and the third sealing portion **350** may be bent to face the side surface of the battery pouch **320**. For example, the second sealing portion **340** and the third sealing portion **350** may be folded in the +z direction based on FIG. **4A**. As will be described later, the first sealing portion **330** may also be folded in the +z direction based on FIG. **4A**, along with the second sealing portion **340** and the third sealing portion **350**. Accordingly, the first sealing portion **330**, the second sealing portion **340**, and the third sealing portion **350**, which are residual portions of the battery pouch **320** remaining after sealing the battery cell **310**, may occupy a reduced space within the electronic device.

[0085] According to an embodiment of the disclosure, as illustrated in FIGS. **4A**, **4B**, and **4C**, the first sealing portion **330** of the battery pouch **320** may be formed in a bent shape and may include at least one groove **331**. In an embodiment of the disclosure, when the battery pouch **320** is viewed

in the +y direction based on FIG. 4B, the first sealing portion 330 may have a shape in which a “.Math.” shape, a “ $\square$ ” shape, and/or a “U” shape are repeatedly formed. Alternatively, the first sealing portion 330 may include at least one of the “.Math.” shape, the “ $\square$ ” shape, and/or the “U” shape. In an embodiment of the disclosure, the groove 331 of the first sealing portion 330 may be a concave space formed as a result of the bending shape of the first sealing portion 330, such as the “.Math.” shape, the “ $\square$ ” shape, and/or the “U” shape.

[0086] In an embodiment of the disclosure, the first sealing portion 330 may have a shape bent by being pressed through a jig (not illustrated). Referring to FIGS. 4A, 4B, and 4C, at least one groove 331 may be formed in the area pressed by the jig. During the sealing process, the first sealing portion 330 may be pressed by placing a jig with a bent shape on the first surface 330A and the second surface 330B of the first sealing portion 330, thereby forming the groove 331 on the first surface 330A and/or the second surface 330B. In addition, the groove 331 may be formed in the first sealing portion 330 through various processes.

[0087] As will be described later, at least a portion of an electronic component disposed in the protection circuit module 400 may be positioned in the groove 331 of the first sealing portion 330. In an embodiment of the disclosure, at least one of the first electronic element 421, the second electronic element 422, the first connection terminal 431, or the second connection terminal 432 of the protection circuit module 400 may be disposed in the groove 331. For example, referring to FIG. 6B, the protection circuit module 400 may be in close contact with the first surface 330A of the first sealing portion 330 such that the first electronic element 421, which is disposed on the first surface 410A of the printed circuit board 410, is positioned in the groove 331 of the first sealing portion 330.

[0088] According to an embodiment of the disclosure, as illustrated in FIGS. 5A and 5B, the first sealing portion 330 may be processed into a bent shape in which the groove 331 is formed and then bent to face the battery cell 310. For example, the first sealing portion 330 may be folded in the +z direction based on FIG. 5B. Accordingly, the space occupied by the first sealing portion 330 in the area where the battery structure 300 is disposed within the electronic device may be reduced. Meanwhile, the first sealing portion 330 may be separated from the second sealing portion 340 and the third sealing portion 350 to allow folding.

[0089] In an embodiment of the disclosure, referring to FIG. 4C, the groove 331 formed in the first sealing portion 330 may be spaced apart by a predetermined distance L1 from an end of the battery pouch 320 that forms the accommodation portion 321, so that the first sealing portion 330 may be folded in the +z direction. Accordingly, the first sealing portion 330 may be bent to face the battery cell 310.

[0090] According to an embodiment of the disclosure, the protection circuit module 400 may be disposed on the first sealing portion 330. In an embodiment of the disclosure, referring to FIGS. 6A and 6B, the protection circuit module 400 may be disposed on the first surface 330A of the first sealing portion 330 in a state in which the first sealing portion 330 is folded in the +z direction based on FIG. 6A to face the side surface of the battery pouch 320. In an embodiment of the disclosure, referring to FIG. 6B, the protection circuit module 400 may be disposed such that the first electronic element 421 is positioned in the groove 331 formed on the first surface 330A of the first sealing portion 330. For example, the printed circuit board 410 may be disposed such that the first surface 410A, on which the first electronic element 421 is disposed, faces the first surface 330A of the first sealing portion 330. The second electronic element 422, which is disposed on the second surface 410B of the printed circuit board 410, may face a counterpart of the electronic device rather than the first sealing portion 330. In this case, the first electronic element 421 may be disposed in the groove 331 of the first sealing portion 330 to be protected from the surrounding environment. Accordingly, the first electronic element 421 may be positioned in the groove 331 of the first sealing portion 330 to be protected from external impact applied to the electronic device. In addition, in the protection circuit module 400, as the first electronic element 421 is positioned in

the groove **331** of the first sealing portion **330**, the space occupied by the first electronic element **421** within the electronic device may be reduced. Accordingly, the size of the battery cell **310** may be increased by the space previously occupied by the first electronic element **421**, thereby securing additional capacity for the battery cell **310**.

[0091] In an embodiment not illustrated in the drawings, the protection circuit module **400** may be disposed in a space defined between the first sealing portion **330** and the accommodation portion **321**. In an embodiment of the disclosure, a groove (not illustrated) may be formed on the second surface **330B** of the first sealing portion **330**. The protection circuit module **400** may be disposed such that either the first electronic element **421** or the second electronic element **422** is positioned in the groove formed on the second surface **330B** of the first sealing portion **330**. In an embodiment of the disclosure, the protection circuit module **400** may be disposed such that the first electronic element **421** is positioned in the groove formed on the second surface **330B** of the first sealing portion **330**. For example, the printed circuit board **410** may be disposed such that the first surface **410A**, on which the first electronic element **421** is disposed, faces the second surface **330B** of the first sealing portion **330**. In some embodiments, the protection circuit module **400** may be disposed such that the second electronic element **422** is positioned in the groove formed on the second surface **330B** of the first sealing portion **330**. For example, the printed circuit board **410** may be disposed such that the second surface **410B**, on which the second electronic element **422** is disposed, faces the second surface **330B** of the first sealing portion **330**. Accordingly, the printed circuit board **410** may be positioned inside the first sealing portion **330**, and the electronic elements **421** and **422** may be positioned in the groove of the first sealing portion **330**, thereby being protected from external impact applied to the electronic device. As either the first electronic element **421** or the second electronic element **422** of the protection circuit module **400** is positioned in the groove of the first sealing portion **330**, the space occupied by the electronic elements **421** and **422** within the electronic device may be reduced. Accordingly, the size of the battery cell **310** may be increased by the space previously occupied by the first electronic elements **421** and **421**, thereby securing additional capacity for the battery cell **310**.

[0092] In an embodiment not illustrated in the drawings, at least one of the first connection terminal **431** and the second connection terminal **432** may be positioned in the groove **331** formed on the first surface **330A** of the first sealing portion **330**. Alternatively, in some embodiments, at least one of the first connection terminal **431** and the second connection terminal **432** may be positioned in a groove formed on the second surface **330B** of the first sealing portion **330**. Accordingly, at least one of the first connection terminal **431** and the second connection terminal **432** may be protected from external impact applied to the electronic device. Furthermore, as either the first connection terminal **431** or the second connection terminal **432** of the protection circuit module **400** is positioned in the groove of the first sealing portion **330**, the space occupied by the connection terminals **431** and **432** within the electronic device may be reduced. Accordingly, the size of the battery cell **310** may be increased by the space previously occupied by the connection terminals **431** and **432**, thereby securing additional capacity for the battery cell **310**.

[0093] Hereinafter, for convenience of explanation, it will be assumed that the protection circuit module **400** is disposed such that the first electronic element **421**, which is disposed on the first surface **410A** of the printed circuit board **410**, is positioned in the groove **331** formed on the first surface **330A** of the first sealing portion **330**, with the first surface **330A** of the first sealing portion **330** facing the first surface **310A** of the first sealing portion **330**.

[0094] Meanwhile, as described above, when the protection circuit module **400** is positioned in the space formed between the first sealing portion **330** and the accommodation portion **321**, the groove **331** formed on the second surface **330B** of the first sealing portion **330** may be additionally spaced apart from an end of the battery pouch **320** that forms the accommodation portion **321** by a predetermined distance **L1** and a distance corresponding to the thickness of the printed circuit board **410** to allow the first sealing portion **330** to be folded in the +z direction of FIG. 4C.

[0095] According to an embodiment of the disclosure, as illustrated in FIGS. 6A and 6B, as the first electronic element **421** of the protection circuit module **400** is positioned in the groove **331** of the first sealing portion **330**, the space occupied by the first electronic element **421** within the electronic device may be reduced. In an embodiment of the disclosure, as the first electronic element **421** is positioned in the groove **331**, the length of the battery structure **300** in the y-axis direction may be reduced based on FIG. 6A. For example, when the groove **331** is not formed in the first sealing portion **330**, the distance from one end of the battery pouch **320** to the end of the second electronic element **422** of the protection circuit module **400** may be about 5.7 mm. In comparison, as illustrated in FIGS. 6A and 6B, when the first electronic element **421** of the protection circuit module **400** is positioned in the groove **331** of the first sealing portion **330**, the distance **L2** from the one end of the battery pouch **320** that forms the accommodation portion **321** to the second electronic element **422** of the protection circuit module **400** may be reduced to about 4.2 mm. In this case, as the space occupied by the first electronic element **421** is reduced through the groove **331** of the first sealing portion **330**, the size of the battery cell **310** may be increased, thereby increasing the capacity of the battery cell **310**.

[0096] In an embodiment of the disclosure, referring to FIGS. 6A and 6B, the protection circuit module **400** may be disposed on the first surface **330A** or the second surface **330B** of the first sealing portion **330** to be electrically connected to the electrode terminal **360** of the battery structure **300**. In an embodiment of the disclosure, at least a section of the first electrode tab **361** of the battery structure **300** may be bent to be connected to the first connection terminal **431** disposed on the printed circuit board **410** of the protection circuit module **400**. Likewise, at least a section of the second electrode tab **362** of the battery structure **300** may be bent to be connected to the second connection terminal **432** disposed on the printed circuit board **410** of the protection circuit module **400**.

[0097] FIGS. 7A and 7B illustrate views for comparing and describing a battery structure with a battery structure according to an embodiment of the disclosure.

[0098] Referring to FIGS. 7A and 7B, the structural differences between a state before the space occupied by the protection circuit module **400** within the space occupied by the battery cell **310** in the electronic device (e.g., the electronic device **101** in FIG. 1) is reduced (e.g., FIG. 7A) and a state after the space occupied by the protection circuit module **400** is reduced (e.g., FIG. 7B) will be described.

[0099] In an embodiment of the disclosure, referring to FIG. 7A, the first sealing portion **330** may be disposed in a direction parallel to the extension direction (e.g., the y-axis direction of FIGS. 7A and 7B) of the battery structure **300** without being bent to face the side surface of the battery pouch **320** of the battery structure **300** (e.g., the battery **189** in FIG. 1). The protection circuit module **400** may be seated on the first sealing portion **330** to be electrically connected to the electrode terminal **360** of the battery structure **300**.

[0100] Referring to FIG. 7B, which illustrates an embodiment of the disclosure, the first sealing portion **330** may be processed into a bent shape in which the groove **331** is formed and may then bent to face the side surface of the battery pouch **320**. For example, the first sealing portion **330** may be folded in the +z direction based on FIG. 7B. In this case, compared to FIG. 7A, the space occupied by the first sealing portion **330** within the electronic device may be reduced. In addition, the protection circuit module **400** may be disposed on the first surface **330A** of the first sealing portion **330** to be electrically connected to the electrode terminal **360** of the battery structure **300**. As the printed circuit board **410** of the protection circuit module **400** is disposed on the first sealing portion **330** such that the first electronic element **421** is positioned in the groove **331** of the first sealing portion **330**, the space occupied by the first electronic element **421** may be reduced. In summary, compared to FIG. 7A, in the embodiment illustrated in FIG. 7B, as the first sealing portion **330** is folded to be oriented in the +Z direction and the first electronic element **421** is positioned in the groove **331** of the first sealing portion **330**, the space occupied by the protection

circuit module **400** within the electronic device may be reduced by Y in FIGS. 7A and 7B.

[0101] In an embodiment of the disclosure, the space Y corresponding to the difference between the space Y1 from one end of the battery pouch **320** to the end of the protection circuit module **400** in FIG. 7A and the space Y2 from one end of the battery pouch **320** to the end of the protection circuit module **400** in FIG. 7B may be secured. Accordingly, the size of the battery cell **310** may be increased by the reduced space Y, thereby securing the capacity of the battery cell **310**. In an embodiment of the disclosure, compared to the battery structure **300** illustrated in FIG. 7A, the battery structure **300** illustrated in FIG. 7B may secure about 5% to 8% additional capacity for the battery cell **310**. The above description may also be applicable to the case where the protection circuit module **400** is positioned in the space formed between the accommodation portion **321** and the first sealing portion **330** to be disposed on the second surface **330B** of the first sealing portion **330**.

[0102] Meanwhile, in the case of a wearable electronic device, its size may be smaller compared to a portable electronic device (e.g., a smartphone or a laptop). Similarly, the battery structure **300** disposed within a wearable electronic device may be relatively smaller than a battery structure **300** used in electronic devices, such as smartphones and laptops. When the battery structure **300** has a small size, such as in a wearable electronic device, even a partial reduction in the space occupied by the first sealing portion **330** and the protection circuit module **400** may result in a relatively high proportion of additional secured space for the increase of the battery cell **310** within the overall size of the battery structure **300**.

[0103] Although the above description assumes that the battery structure **300** is used in a wearable electronic device, the disclosure may not be limited thereto. For example, the battery structure **300** of the disclosure may be applied to various types of electronic devices, such as laptops, smartphones, or tablet PCs.

[0104] According to an embodiment of the disclosure, an electronic device **101** may include a battery cell **310**, a battery pouch **320** sealing the battery cell, the battery pouch including an accommodation portion **321** in which the battery cell is disposed and a first sealing portion **330** that is bent to face the battery cell and includes at least one groove **331**, an electrode terminal **360** including a first electrode tab **361** connected to a positive electrode plate of the battery cell and drawn out to the first sealing portion and a second electrode tab **362** connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion, and a protection circuit module **400** (e.g., the battery protection circuit **240**) including at least one electronic element **420** and disposed on the first sealing portion such that the electronic element is positioned in the groove of the first sealing portion, the protection circuit module being connected to the electrode terminal.

[0105] The protection circuit module may include a printed circuit board **410** on which the electronic element is disposed, and a connection terminal **430** disposed on the printed circuit board. The connection terminal may include a first connection terminal **431** connected to the first electrode tab and a second connection terminal **432** connected to the second electrode tab.

[0106] The printed circuit board may be disposed on a first surface **330A** of the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion.

[0107] The printed circuit board may be positioned in a space defined between the first sealing portion and the accommodation portion, and may be disposed on a second surface **330B** of the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion.

[0108] The battery pouch may include a second sealing portion **340** that is positioned in a first direction with respect to the battery cell (e.g., the +x direction based on FIG. 3) and is bent to face the battery cell, and a third sealing portion (**350**) that is positioned in a second direction with respect to the battery cell (e.g., the -X direction with based on FIG. 3) and is bent to face the battery cell.

[0109] The first sealing portion, the second sealing portion, and the third sealing portion may be

bent in the same direction.

[0110] The groove of the first sealing portion may have a size including the electronic element of the protection circuit module.

[0111] A section of the first electrode tab of the electrode terminal may be bent so as to be connected to the first connection terminal of the printed circuit board, and a section of the second electrode tab of the electrode terminal may be bent so as to be connected to the second connection terminal of the printed circuit board.

[0112] The groove of the first sealing portion may be formed at a predetermined distance **L1** from an end of the battery pouch that defines the accommodation portion, such that the first sealing portion is bendable with respect to the battery cell.

[0113] In addition, the battery cell may be a lithium battery.

[0114] According to an embodiment of the disclosure, a battery structure **300** (e.g., the battery cell **189**) may include a battery cell **310**, a battery pouch **320** sealing the battery cell, the battery pouch including an accommodation portion **321** in which the battery cell is disposed and a first sealing portion **330** that is bent to face the battery cell and includes at least one groove **331**, an electrode terminal **360** including a first electrode tab **361** connected to a positive electrode plate of the battery cell and drawn out to the first sealing portion and a second electrode tab **362** connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion, and a protection circuit module **400** (e.g., the battery protection circuit **240**) including at least one electronic element **420** and disposed on the first sealing portion such that the electronic element is positioned in the groove of the first sealing portion, the protection circuit module being connected to the electrode terminal.

[0115] The protection circuit module may include a printed circuit board **410** on which the electronic element is disposed, and a connection terminal **430** disposed on the printed circuit board. The connection terminal may include a first connection terminal **431** connected to the first electrode tab and a second connection terminal **432** connected to the second electrode tab.

[0116] The printed circuit board may be disposed on a first surface **330A** of the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion.

[0117] The printed circuit board may be positioned in a space defined between the first sealing portion and the accommodation portion, and may be disposed on a second surface **330B** of the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion.

[0118] The battery pouch may include a second sealing portion **340** that is positioned in a first direction with respect to the battery cell (e.g., the +x direction based on FIG. 3) and is bent to face the battery cell, and a third sealing portion (**350**) that is positioned in a second direction with respect to the battery cell (e.g., the -x direction based on FIG. 3) and is bent to face the battery cell.

[0119] The first sealing portion, the second sealing portion, and the third sealing portion may be bent in the same direction.

[0120] The groove of the first sealing portion may have a size including the electronic element of the protection circuit module.

[0121] A section of the first electrode tab of the electrode terminal may be bent so as to be connected to the first connection terminal of the printed circuit board, and a section of the second electrode tab of the electrode terminal may be bent so as to be connected to the second connection terminal of the printed circuit board.

[0122] The groove of the first sealing portion may be formed at a predetermined distance **L1** from an end of the battery pouch that defines the accommodation portion, such that the first sealing portion is bendable with respect to the battery cell.

[0123] In addition, the battery cell may be a lithium battery.

[0124] According to an embodiment of the disclosure, a space between components constituting the battery structure **300** may be reduced. In this case, the capacity of the battery cell **310** may be

increased by expanding the size of the battery cell **310** in proportion to the reduced space between the components. For example, the space occupied by the battery cell **310** protection circuit module **400** in a battery cell **310** disposition space may be reduced, and the size of the battery cell **310** may be increased by the reduced space.

[0125] It will be appreciated that various embodiments of the disclosure according to the claims and description in the specification can be realized in the form of hardware, software or a combination of hardware and software.

[0126] Any such software may be stored in non-transitory computer readable storage media. The non-transitory computer readable storage media store one or more computer programs (software modules), the one or more computer programs include computer-executable instructions that, when executed by one or more processors of an electronic device, cause the electronic device to perform a method of the disclosure.

[0127] Any such software may be stored in the form of volatile or non-volatile storage, such as, for example, a storage device like read only memory (ROM), whether erasable or rewritable or not, or in the form of memory, such as, for example, random access memory (RAM), memory chips, device or integrated circuits or on an optically or magnetically readable medium, such as, for example, a compact disk (CD), digital versatile disc (DVD), magnetic disk or magnetic tape or the like. It will be appreciated that the storage devices and storage media are various embodiments of non-transitory machine-readable storage that are suitable for storing a computer program or computer programs comprising instructions that, when executed, implement various embodiments of the disclosure. Accordingly, various embodiments provide a program comprising code for implementing apparatus or a method as claimed in any one of the claims of this specification and a non-transitory machine-readable storage storing such a program.

[0128] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a battery cell; a battery pouch sealing the battery cell, the battery pouch comprising an accommodation portion in which the battery cell is disposed and a first sealing portion that is bent to face the battery cell and comprises at least one groove; an electrode terminal comprising a first electrode tab connected to a positive electrode plate of the battery cell and drawn out to the first sealing portion and a second electrode tab connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion; and a protection circuit module comprising at least one electronic element and disposed on the first sealing portion such that the electronic element is positioned in the groove of the first sealing portion, wherein the protection circuit module is connected to the electrode terminal.
2. The electronic device of claim 1, wherein the protection circuit module comprises: a printed circuit board on which the electronic element is disposed, and a connection terminal disposed on the printed circuit board, and wherein the connection terminal comprises: a first connection terminal connected to the first electrode tab, and a second connection terminal connected to the second electrode tab.
3. The electronic device of claim 2, wherein the printed circuit board is disposed on a first surface of the first sealing portion such that the electronic element is positioned in the groove of the first sealing portion.
4. The electronic device of claim 2, wherein the printed circuit board is positioned in a space defined between the first sealing portion and the accommodation portion, and wherein the printed circuit board is disposed on a second surface of the first sealing portion, such that the electronic

element is positioned in the groove of the first sealing portion.

5. The electronic device of claim 1, wherein the battery pouch comprises: a second sealing portion that is positioned in a first direction with respect to the battery cell and is bent to face the battery cell; and a third sealing portion that is positioned in a second direction with respect to the battery cell and is bent to face the battery cell.

6. The electronic device of claim 5, wherein the first sealing portion, the second sealing portion, and the third sealing portion are bent in a same direction.

7. The electronic device of claim 1, wherein the groove of the first sealing portion has a size comprising the electronic element of the protection circuit module.

8. The electronic device of claim 2, wherein a section of the first electrode tab of the electrode terminal is bent so as to be connected to the first connection terminal of the printed circuit board, and wherein a section of the second electrode tab of the electrode terminal is bent so as to be connected to the second connection terminal of the printed circuit board.

9. The electronic device of claim 1, wherein the groove of the first sealing portion is formed at a predetermined distance (L1) from an end of the battery pouch that defines the accommodation portion, such that the first sealing portion is bendable with respect to the battery cell.

10. The electronic device of claim 1, wherein the battery cell is a lithium-ion battery.

11. A battery structure comprising: a battery cell; a battery pouch sealing the battery cell, the battery pouch comprising an accommodation portion in which the battery cell is disposed and a first sealing portion that is bent to face the battery cell and comprises at least one groove; an electrode terminal comprising a first electrode tab connected to a positive electrode plate of the battery cell and drawn out to the first sealing portion and a second electrode tab connected to a negative electrode plate of the battery cell and drawn out to the first sealing portion; and a protection circuit module comprising at least one electronic element and disposed on the first sealing portion such that the electronic element is positioned in the groove of the first sealing portion, wherein the protection circuit module is connected to the electrode terminal.

12. The battery structure of claim 11, wherein the protection circuit module comprises: a printed circuit board on which the electronic element is disposed, and a connection terminal disposed on the printed circuit board, and wherein the connection terminal comprises: a first connection terminal connected to the first electrode tab, and a second connection terminal connected to the second electrode tab.

13. The battery structure of claim 12, wherein the printed circuit board is disposed on a first surface of the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion.

14. The battery structure of claim 12, wherein the printed circuit board is positioned in a space defined between the first sealing portion and the accommodation portion, and wherein the printed circuit board is disposed on a second surface of the first sealing portion, such that the electronic element is positioned in the groove of the first sealing portion.

15. The battery structure of claim 11, wherein the battery pouch comprises: a second sealing portion that is positioned in a first direction with respect to the battery cell and is bent to face the battery cell; and a third sealing portion that is positioned in a second direction with respect to the battery cell and is bent to face the battery cell.

16. The battery structure of claim 15, wherein the first sealing portion, the second sealing portion, and the third sealing portion are bent in a same direction.

17. The battery structure of claim 11, wherein the groove of the first sealing portion has a size comprising the electronic element of the protection circuit module.

18. The battery structure of claim 12, wherein a section of the first electrode tab of the electrode terminal is bent so as to be connected to the first connection terminal of the printed circuit board, and wherein a section of the second electrode tab of the electrode terminal is bent so as to be connected to the second connection terminal of the printed circuit board.

- 19.** The battery structure of claim 11, wherein the groove of the first sealing portion is formed at a predetermined distance (L1) from an end of the battery pouch that defines the accommodation portion, such that the first sealing portion is bendable with respect to the battery cell.
- 20.** The battery structure of claim 11, wherein the battery cell is a lithium-ion battery.
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